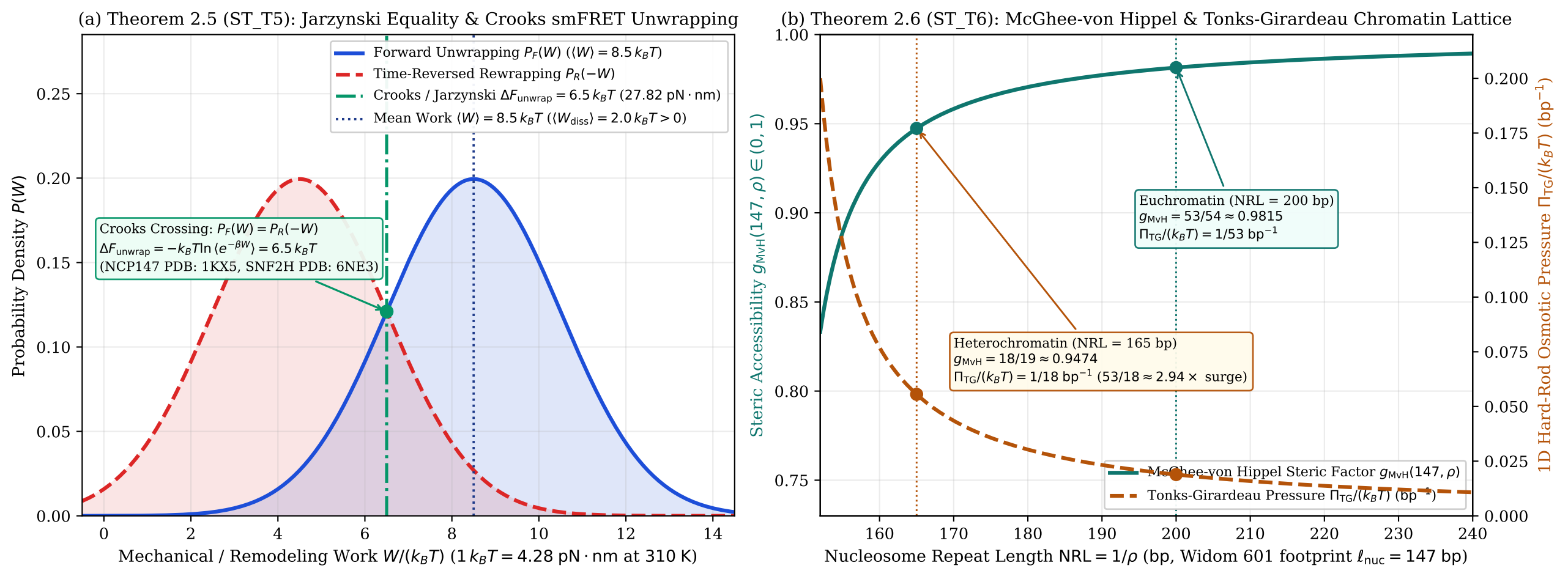

(a) Theorem 2.5 (ST_T5): Jarzynski Equality & Crooks smFRET Unwrapping
Forward Unwrapping $P_F(W)$ ($\langle W\rangle = 8.5\,k_BT$)
Time-Reversed Rewrapping $P_R(-W)$
Crooks / Jarzynski $\Delta F_{\text{unwrap}} = 6.5\,k_BT$ (27.82 pN · nm)
Mean Work $\langle W\rangle = 8.5\,k_BT$ ($\langle W_{\text{diss}}\rangle = 2.0\,k_BT > 0$)
Crooks Crossing: $P_F(W) = P_R(-W)$
$\Delta F_{\text{unwrap}} = -k_BT \ln\langle e^{-\beta W}\rangle = 6.5\,k_BT$
(NCP147 PDB: 1KX5, SNF2H PDB: 6NE3)
Probability Density $P(W)$
Mechanical / Remodeling Work $W/(k_BT)$ ($1\,k_BT = 4.28$ pN · nm at 310 K)
(b) Theorem 2.6 (ST_T6): McGhee-von Hippel & Tonks-Girardeau Chromatin Lattice
Euchromatin (NRL = 200 bp)
$g_{\text{MvH}} = 53/54 \approx 0.9815$
$\Pi_{\text{TG}}/(k_BT) = 1/53$ bp^{-1}
Heterochromatin (NRL = 165 bp)
$g_{\text{MvH}} = 18/19 \approx 0.9474$
$\Pi_{\text{TG}}/(k_BT) = 1/18$ bp^{-1} ($53/18 \approx 2.94\times$ surge)
McGhee-von Hippel Steric Factor $g_{\text{MvH}}(147, \rho)$
Tonks-Girardeau Pressure $\Pi_{\text{TG}}/(k_BT)$ (bp^{-1})
Steric Accessibility $g_{\text{MvH}}(147, \rho) \in (0, 1)$
1D Hard-Rod Osmotic Pressure $\Pi_{\text{TG}}/(k_BT)$ (bp^{-1})
Nucleosome Repeat Length NRL $= 1/\rho$ (bp, Widom 601 footprint $\ell_{\text{nuc}} = 147$ bp)